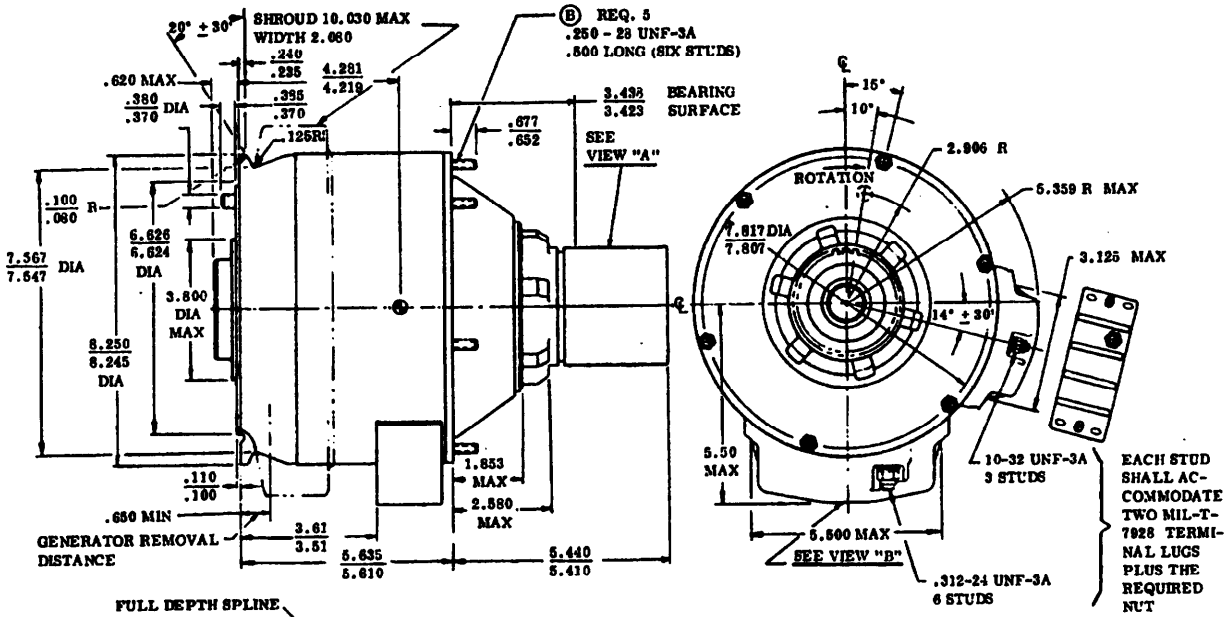
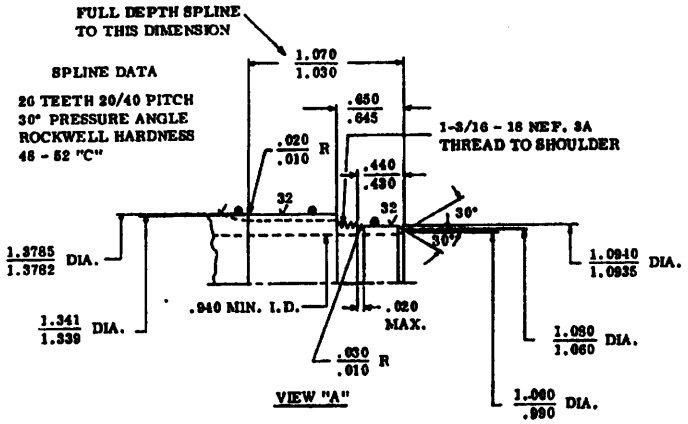
MIL-G-21480/10B(AS)
20 December 1976
SUPERSEDES
MIL-G-21480/10A(AS)
1 DEC. 1969

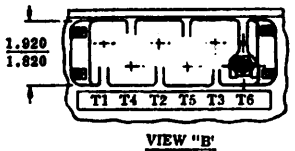
MILITARY SPECIFICATION SHEET
GENERATOR SYSTEM, 20 KVA, 400 HERTZ ALTERNATING CURRENT, STATICALLY EXCITED, BRUSH TYPE

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of MIL-G-21480.







VIEW "B"

DIA MARKED ⊕ TO BE CONCENTRIC WITH DIA MARKED ⊗ WITHIN .005 T. I. R.

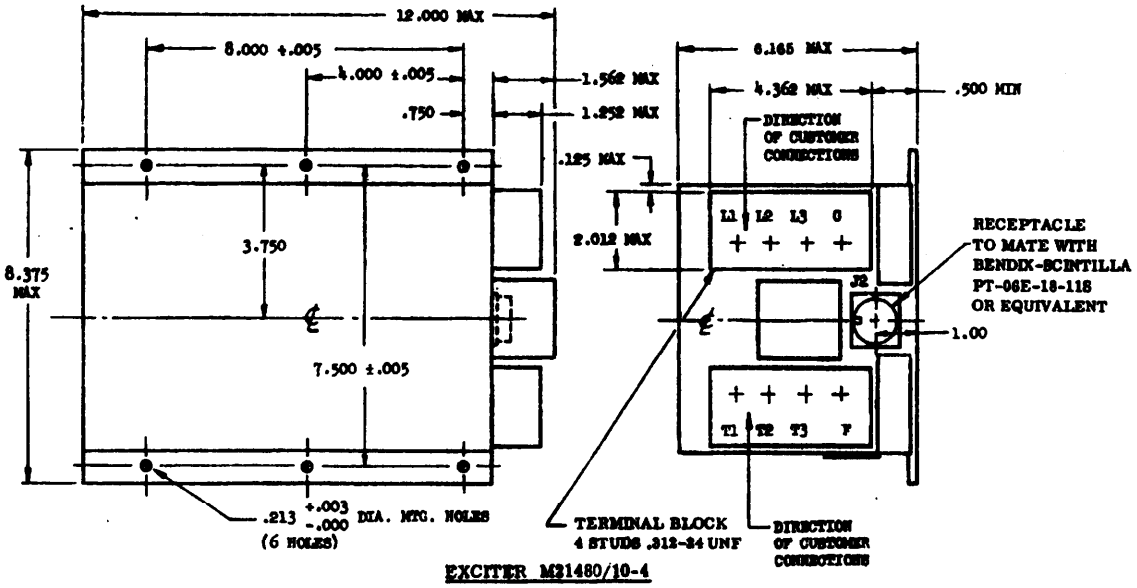
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Ⓑ DENOTES CHANGES

GENERATOR MS1480/10-1

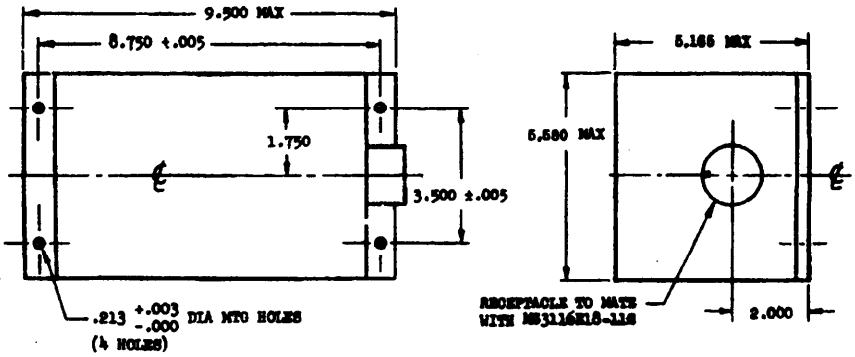
FSC 6115

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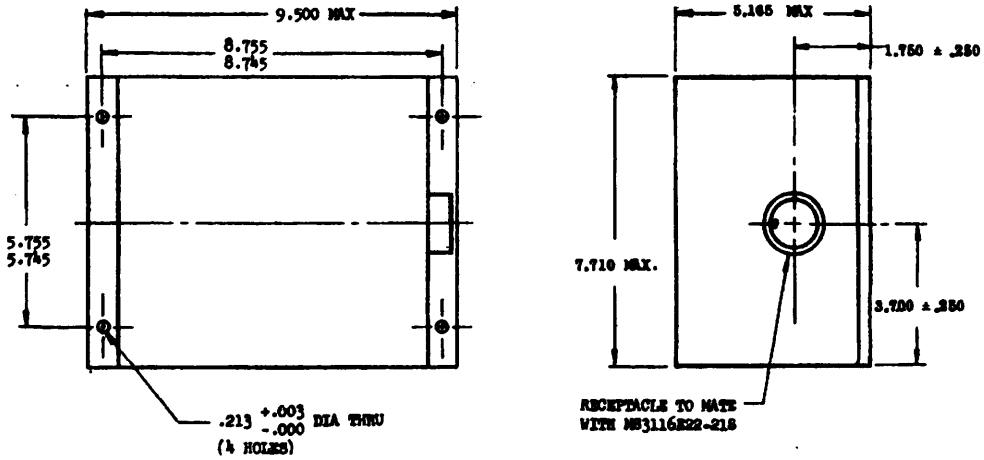
EXCITER M21480/10-4

MAXIMUM WEIGHT 16.9 POUNDS



REGULATOR M21480/10-2

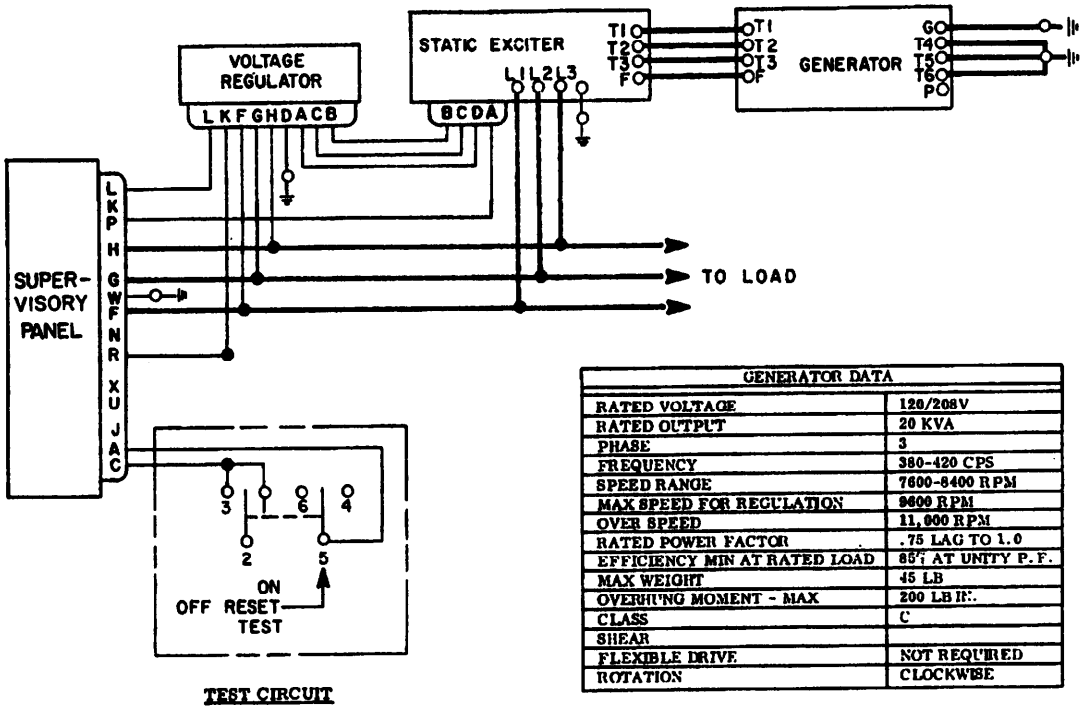
MAXIMUM WEIGHT 6.25 POUNDS



SUPERVISORY PANEL M21480/10-3

MAXIMUM WEIGHT 9.1 POUNDS

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TEST CIRCUIT

GENERATOR DATA	
RATED VOLTAGE	120/208V
RATED OUTPUT	20 KVA
PHASE	3
FREQUENCY	380-420 CPS
SPEED RANGE	7600-8400 RPM
MAX SPEED FOR REGULATION	8400 RPM
OVER SPEED	11,000 RPM
RATED POWER FACTOR	.75 LAG TO 1.0
EFFICIENCY MIN AT RATED LOAD	85% AT UNITY P.F.
MAX WEIGHT	45 LB
OVERHUNG MOMENT - MAX	200 LB IN.
CLASS	C
SHEAR	
FLEXIBLE DRIVE	NOT REQUIRED
ROTATION	CLOCKWISE

NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances: decimals ± .031, angles ± 1°.

(B) REQUIREMENTS:

1. Where only maximum dimensions are shown, the part need not have the shape shown but should be contained within the outline shown. Protrusions such as rivet and screw heads shall be contained within the outlines shown.

2. Flight test: 100 hours flight time on an A-6 Aircraft is required for qualification.

3. Generator shall be finished in a color conforming to FED-STD-595 Color No. 17038, 27038, or 37038.

4. For mounting purposes, supply a water-proofed package of six (6) MS-90415-4 nuts securely attached to each generator assembly.

(B) 5. Flight maneuver forces - The components and their mounting provisions shall withstand the following conditions without permanent deformation or failure:

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(1) Linear acceleration (acting at the center of gravity of the unit)

(a) 7.0g's acting vertically upward or downward (along Z-axis)

(b) 5.0g's acting forward and 3g's aft (along X-axis)

(c) 2.0g's acting sideward in either direction (along Y-axis)

(2) The components and their mounting provisions shall not fail when subjected to static loads equivalent to 1.5 times the loads specified in (1) above.

(B) 6. Identification of forces - The direction of the forces quoted herein shall be identified by the following axes:

Unit Orientation (Direction)	Corresponding Unit Axis
Longitudinal (fore or aft)	X
Transverse (sideward)	Y
Downward or upward	Z

The unit is in the normal horizontal position when the Z-axis coincides with the direction of gravity.

(B) 7. Gyroscopic moments - At maximum in-flight speed, the components and their mountings shall withstand a gyroscopic moment imposed by steady angular velocities, as tabulated in Table 1, for a period of 30 seconds.

TABLE 1

Axis	Angular Limit (Degrees)	Velocity (Degrees/Second)	Acceleration (Rad/Sec^2) Landing	In-Flight
Pitch	None (360)	50	15	7
Roll	None (360)	125	15	30
Yaw	± 15	20	15	0.5

(B) 8. Load capacity - With blast cooling as specified in 3.4.9.6.1 at altitudes up to 65,000 feet, and at a power factor of 0.75 to 0.90 lagging, the generator shall be capable of supplying an overload of 30 kva for a period of two minutes and an overload of 40 kva at 0.75 for a period of five seconds. With self-induced cooling, at altitudes up to 6000 feet, and at a power factor of 0.75 lagging, only the 40 kva overload for five seconds shall be required. The voltage and frequency shall remain within the steady state limits of 3.5 after the initial transients for the load conditions described herein.

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(B) 9. Generator speed - The generator system shall be required to supply the ac power output specified herein, with the following input speed conditions supplied by the procuring agency constant speed drive unit.

(B) 10. Speed regulation - The constant speed drive unit shall be capable of maintaining speed regulation within the following speed ranges and under the usage conditions specified below, when supplied with input conditions specified for the constant speed drive. The specified frequency control tolerances for these speed ranges and imposed transients are for the operation of the unit under regulation of a centrifugally actuated pneumatic speed sensor. The unit shall be capable of maintaining normal rated speed within ± 1.0 percent under the control of the centrifugally actuated pneumatic speed sensor, provided the inlet air, input shaft speed, and generator loads are maintained under steady-state conditions. The transient frequency deviation limitations specified in 3.9.5 shall be maintained by the pneumatic control.

(B) 11. Engine-driven and ground checkout steady-state speed regulation - For steady-state conditions of load input speed as permitted by the constant speed drive requirements, the unit shall provide a constant output frequency of 400 ± 4.0 cps when the unit is engine-driven over an input drive pad steady-state speed range between zero and 7000 rpm. Input steady-state speed is defined as engine drive pad input speed with accelerations not exceeding ± 10 rpm per second. Steady-state load is defined as the output power variation not exceeding ± 1 kW.

(B) 12. Transient frequency control characteristics - When the unit is being driven at engine input speeds within the range of zero to 7000 rpm and with a generator load of 20 kW imposed, the generator speed for conditions of input speed acceleration and deceleration shall be as shown in Table 2 below.

TABLE 2

Input Speed (rpm)	Input Speed Acceleration or Deceleration (rpm per sec)	Drive Output Frequency (cps)
3750-7000	0-50	400 ± 4
3750-7000	50-600	400 ± 5
3750-7000	600-2000	400 ± 20

(B) 13. Cyclic oscillation - There shall be no continuous cyclic oscillation of generator speed in excess of ± 1.00 cps.

(B) 14. Sudden load change - With constant engine input speed, sudden application or removal of load up to the rated power output shall not cause a generator speed deviation in excess of ± 20 cps.

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(B) 15. Combination of transients - A combination of shock load and input speed acceleration or deceleration shall not cause a deviation of output speed greater than the sum of the limits allowed for the individual conditions. The output speed shall return to steady-state limits within two seconds after application of a 20 kW shock load.

(B) 16. Frequency limits - The frequency shall remain within the limits of 396 to 404 cycles per second for all environmental and steady state load conditions. The transient frequency limits for application of 200%, .75 lagging power factor load and for removal of 100%, .75 lagging power factor load shall be 380 to 420 cycles per second. The transient frequency shall return to and remain within the steady state limits within 2 seconds.

(B) 17. Amplitude - The frequency modulation shall be contained within a 4-cycle per second band. This band shall always be within the steady state frequency limits of paragraph 3.9.7.1.

(B) 18. Rate of frequency change - The rate of frequency change due to frequency modulation shall not be greater than 25 cps per seconds.

(B) 19. Short-circuit capacity - The generator and associated regulator and exciter shall be capable of delivering 300 percent rated current at rated speed to a three-phase short circuit for a minimum of five seconds when the generator and regulator are hot as a result of running at rated load. The short-circuit current requirements shall be 500 percent for a line-to-neutral or 300 percent for a line-to-line short applied to any phase.

(B) 20. Efficiency - The efficiency of the generator, as determined at rated speed, from 50 percent to 150 percent rated load and 1.00 power factor, shall meet the requirements of Figure 1. Losses due to the static exciter and voltage regulator shall be considered in the generator efficiency. Control panel losses shall not exceed 75 watts total, under normal operating conditions.

(B) 21. Explosion proof - The static exciter, supervisory panel and voltage regulator shall be subjected to tests of MIL-E-5272, Procedure IV. Items meeting conditions of MIL-E-5272, paragraph 5.4.4 require no tests.

The requirements of MIL-G-21480(AS) will pertain except for the following additions, deletions or exceptions.

(B) DELETE: Paragraphs 3.4.9.2, 3.4.9.5, 4.5.26.

(B) 3.4.8 Rating - Delete and substitute: The generator and its control system shall be rated for operation at 20 kva (8000 rpm nominal), at all power factors from 1.00 to 0.75 lagging. The nominal output of the generator shall be 120/208 volts, 400 cps, three-phase, four-wire, grounded neutral.

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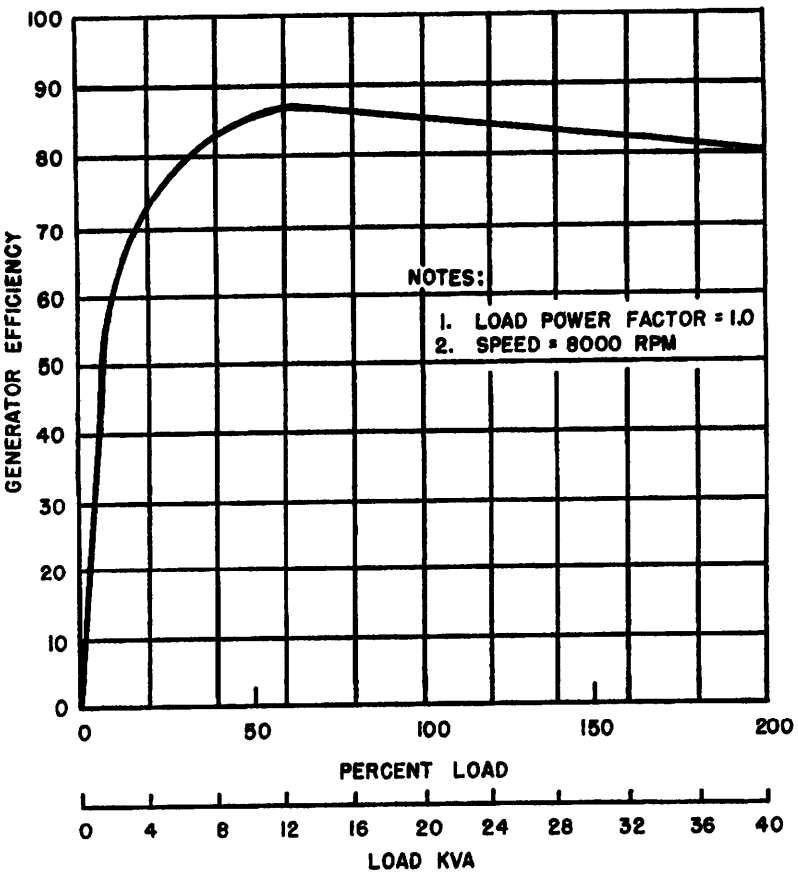
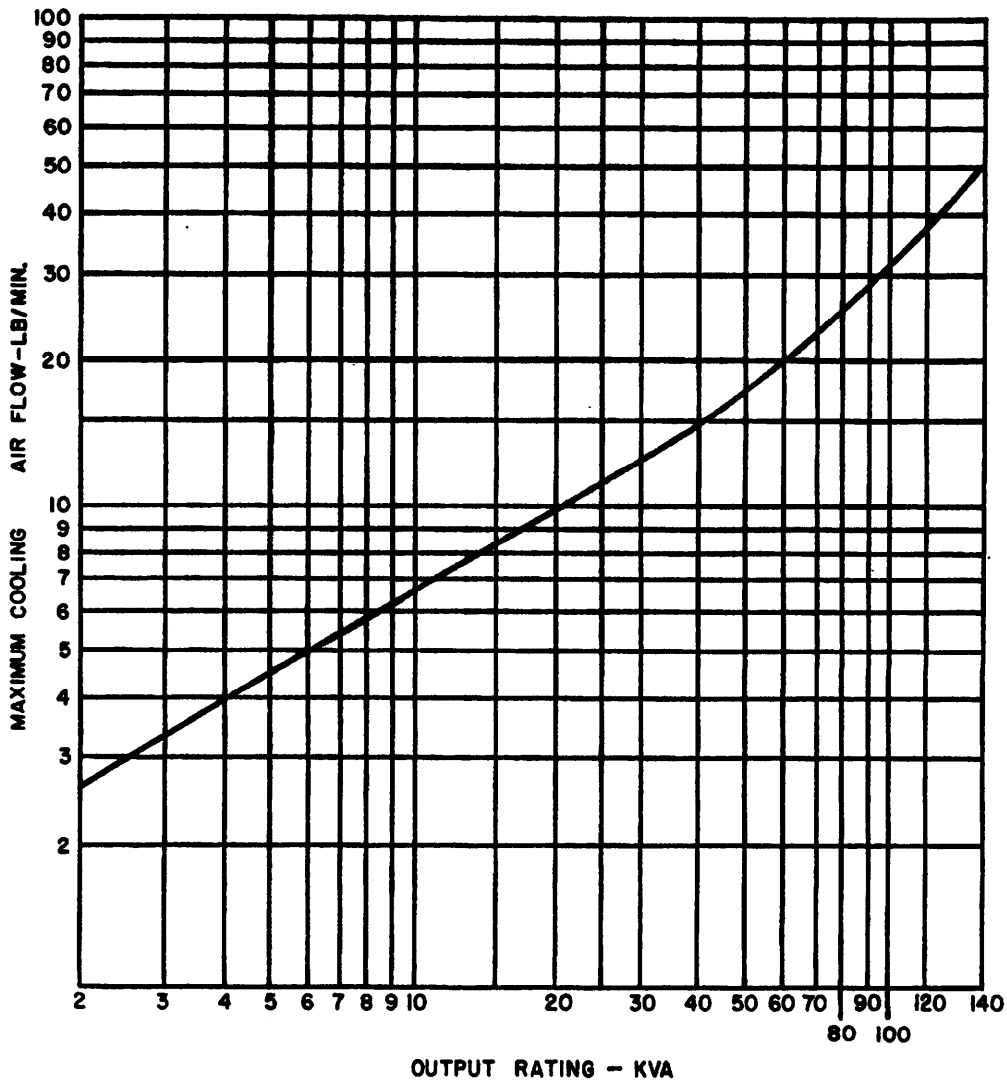


FIGURE 1

(B) 3.4.9.6 Cooling - Add: A ten-inch diameter exhaust shroud having a maximum weight of 13 ounces shall be supplied with the generator.

(B) 3.4.9.6.1 (a) Air cooling - Add: The generator ratings of 3.4.8, 3.4.8.1 and 3.4.8.2 shall be met for continuous operation with the exhaust shroud specified in 3.4.9.6 removed. The shroud shall be designed to cause no more than a 6 percent loss in generator output capability. As a design objective the generator rating with the exhaust shroud in place shall be that specified in 3.4.8, 3.4.8.1 and 3.4.8.2. The maximum cooling air flow shall be in accordance with Figure 2.

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(B) FIGURE 2. Curve for determination of basic cooling airflow for designed kva output.

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ADD: (b) Self-induced cooling - The generator rating at any altitude up to 6000 feet shall be met as specified in 3.4.8, 3.4.8.1 and 3.4.8.2 with the exhaust shroud removed. The inlet duct shall be so designed that the pressure drop in the ducting for the flow rate shown on Figure 2 shall be 4.0 ± 1/4 in. water, and the inlet air temperature shall be 40° ± 2°C. With these conditions the generator shall be operated until the winding temperature stabilizes. Note: The generator shall be considered to have reached stabilized conditions when the operating temperature of components, such as field windings, does not increase more than 1°C with respect to inlet air temperature during a period of 5 minutes. The inlet air temperature at the various altitudes shall be considered to be in accordance with MIL-STD-210 for hot atmosphere conditions. As a design objective the generator rating with the exhaust shroud in place shall be that specified in 3.4.8, 3.4.8.1 and 3.4.8.2.

(B) ADD: 3.4.9.9 Leakage - The generator shall perform satisfactorily and suffer no adverse effects from lubricating oil in accordance with MIL-L-7808 or MIL-L-23699 that may reach the generator through its seal. The seal will not permit oil leakage in excess of 1.0 cc per hour at ambient pressure.

(B) ADD: 3.4.10.1.1 Function - The generator voltage regulator shall be of the static type, designed to operate with the static exciter of 3.4.10.1.3. The voltage regulator shall be a separately packaged component.

(B) ADD: 3.4.10.1.2 Voltage regulator - The regulator shall incorporate a high phase take-over feature whose take-over voltage shall be such as to prevent an overvoltage from existing due to unbalanced conditions. The regulator shall regulate the line-to-neutral voltages to the following extent:

(a) Steady-state voltage regulation, fixed ambient, no load to full-rated load equals plus one, minus zero percent.

(b) Steady-state voltage regulation, any ambient no load to full-rated load equals ± 2 percent.

(c) Steady-state voltage regulation, any ambient, no load to 200 percent rated load equals ± 2.5 percent.

The point of regulation is that point at which the regulator senses and establishes system voltage.

(B) ADD: 3.4.10.1.3 Static exciter - The static-exciter shall meet the following requirements:

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(B) ADD: 3.4.10.1.3.1 Function - Excitation for the generator shall be provided by a static type exciter which can be located remotely from the generator.

(B) ADD 3.4.10.1.3.2 Contents of unit - The exciter shall contain provisions for maintenance of excitation during faults, overloads, and large, low-power-factor loads. The system shall be self-sufficient and dependent only upon its own power generation for build-up, recovery, and control of voltage. The generator control (de-energizing) relay may be located in the exciter.

3.5.5 - Add: Twenty watts at 28 volt nominal of rectified dc power shall be available for switching on control external to this system.

(B) 3.5.7.2 Overvoltage - Delete "C" comment on Figure 3.

(B) 3.5.7.3 Undervoltage protection - Delete and substitute: The generating system shall provide for the generator to be disconnected from the load bus when any individual phase voltage drops to or below 104 volts, and remains at this limit for a period of from five to seven seconds. The device shall have a minimum pickup of 111 volts. The voltages specified herein are based on individual RMS line to neutral voltages.

(B) 3.5.7.6 Feeder fault - Delete and add: Protection inherent since generating system will collapse on low resistance feeder faults between the generator and the static exciter. Feeder fault protection is not required between the static exciter and the line contactor.

4.5.4 Overspeed - Delete and substitute: The generator rotor shall have the structural capacity of being rotated up to a maximum speed of 11,000 rpm. Operation for 30 seconds at 9000 rpm for 600 times (the anticipated number of engine starts during 1000 hours of operating life) shall not adversely affect the life or operation of the generator. The generator will not be normally energized at speeds above 8400 rpm. The stator shall contain the rotor parts in the event the overspeed circuit malfunctions. The overspeed switch on the drive will be set to trip at 9600 rpm to disconnect the electrical control panel from the system.

4.5.20 Salt spray - Delete first sentence and substitute: The regulator, supervisory panel and static exciter shall be subjected to 50 hours of the Salt spray test outlined in MIL-E-5272.

4.5.25 Humidity - Delete first sentence and substitute: The generator, regulator, supervisory panel, and static exciter shall be subjected to the Humidity Tests of MIL-E-5272, Procedure I.

(B) ADD: 4.5.30 Acceleration - The exciter, regulator and control panel shall be tested in accordance with Procedure III of MIL-E-5272. The magnitude of the acceleration shall be 15 g's.

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DOCUMENT IDENTIFIER AND TITLE
MIL-G-21480/10B(AS) Generator, Systems, 20KVA, 400 Hertz Alternating Current Statically Excited, Brush Type

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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